the nearby periosteum and endosteum and begin reconstructing of the bone. The fibroblasts produce fibres. The chondroblasts secrete the cartilage matrix. Within this repair tissue, osteoblasts begin forming spongy bone. The cartilage matrix later calcifies and forms the fibrocartilaginous callus.

**3. Formation of Bony callus**

New bone trabeculae begin to appear in the fibro cartilaginous callus. Gradually that is converted into a bony (hard) callus of spongy bone. Bony callus formation continues until a firm union is formed about two months later to an year for complete woven bone formation.

**4. Remodeling of Bone**

Bony callus formation will be continued for several months. After that the bony callus is remodelled. The excess material on the diaphysis exterior and within the medullary cavity is removed and the compact bone is laid down to reconstruct the shaft walls. The final structure of the remodelled area resembles like the unbroken bony region.

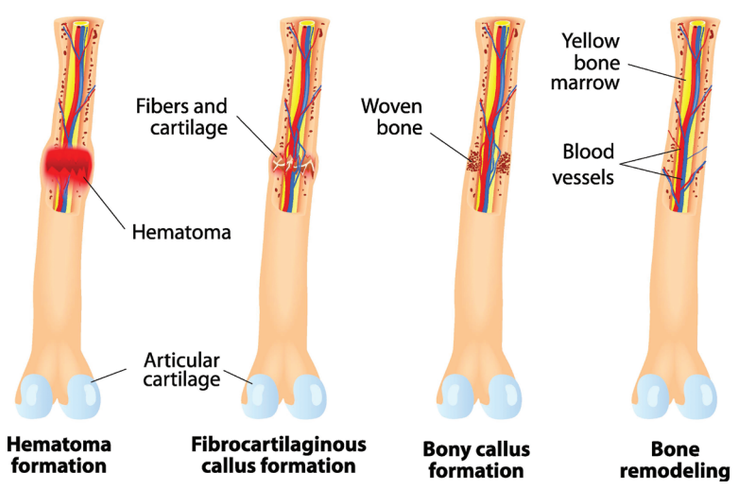



**Figure 9.13** Mechanism and healing of bone fracture

## 9.15 Dislocation of joints and treatment

Dislocation of joint is the total displacement of the articular end of the bone from the joint cavity. The normal alignment of the bones becomes altered. Joints of the jaw, shoulders, fingers and thumbs are most commonly dislocated.

Dislocations of joints are classified as (i) Congenital deformities (ii) Traumatic (iii) Pathological and (iv)Paralytic.

**(i)** **Congenital deformities** are due to genetic factors or factors operating on the developing foetus.

**(ii)** **Traumatic dislocation** is due to serious violence. It occurs in the shoulder, elbow and hip.

**(iii)** **Pathological dislocation** is caused by some diseases like tuberculosis. It may cause dislocation of the hip.

**(iv)** **Paralytic dislocation** caused by paralysis of one group of muscles of an extremity.

### Treatment

If the joint doesn't return to normal condition naturally the following treatments should be given.

- Manipulation or repositioning
- Immobilization
- Medication
- Rehabilitation

## 9.16 Physiotherapy

Physiotherapy is the therapeutic exercise to make the limbs work near normally. It is a rehabilitation profession with a presence in all health



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